

DAEN/ISEN 427 Homework 3

Due date: Tue Sep 23, 2025 (before class)

Rubric: Full credit may be given for any reasonable Bayesian analysis.

- Consider an object moving in a plane with noisy location measurements of the coordinates. Specifically let the state be the coordinates and velocities $s_t^\top := [x \ y \ \dot{x} \ \dot{y}]$.

The generative model is: $s_{t+1} = As_t + w_{t+1}$, where: $w_t \sim \mathcal{N}(0, Q)$ and

$$A = \begin{pmatrix} 1 & 0 & dt & 0 \\ 0 & 1 & 0 & dt \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad Q = \frac{1}{1000} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and the observations $z_t \in \mathbb{R}^2$ are $z_t = Gs_t + v_t$, $v_t \sim \mathcal{N}(0, R)$ where:

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad R = \frac{1}{1000} \begin{pmatrix} 0.2 & -0.05 \\ -0.05 & 0.1 \end{pmatrix}$$

Assume $dt = 0.01$ and $s_0 \sim \mathcal{N}(\hat{s}_0, \Sigma)$ with $\hat{s}_0^\top := [1 \ 1 \ -25 \ -15]$ and $\Sigma = \frac{1}{100}I$.

- Simulate $T = 100$ transitions and observations of the system. Plot the obtained path as well as the Kalman filter mean prediction.

Answer: See figure 1.

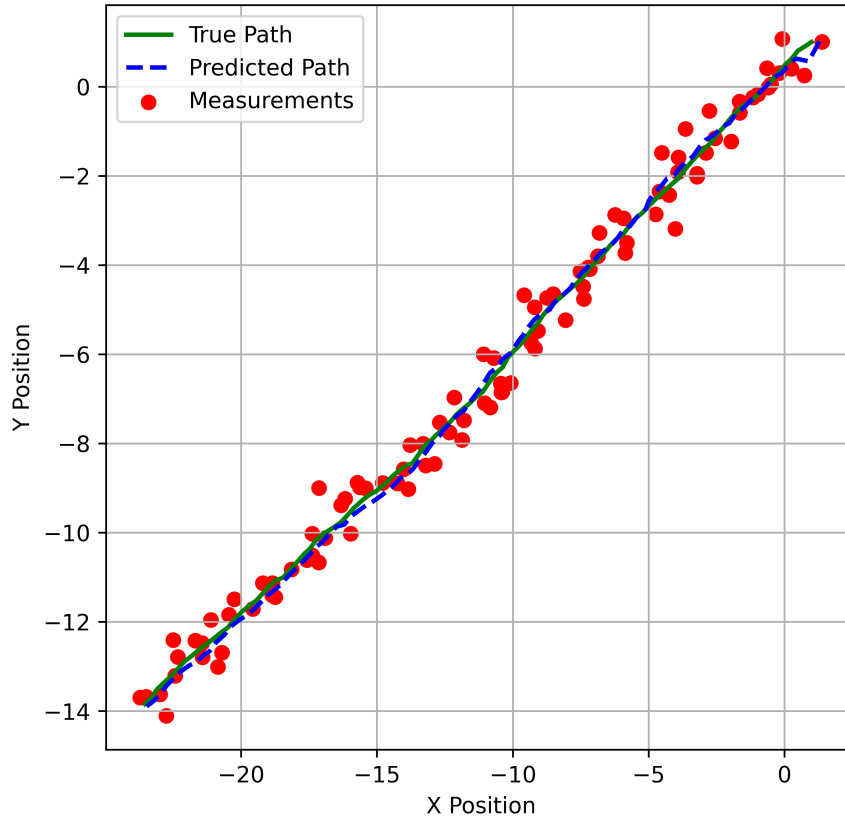


Figure 1: Sample Path and Kalman Prediction

- b. Repeat (a) $N = 100$ times. Compute (and plot) the mean and variance of the trajectories and Kalman filter prediction.

Answer: See figure 2.

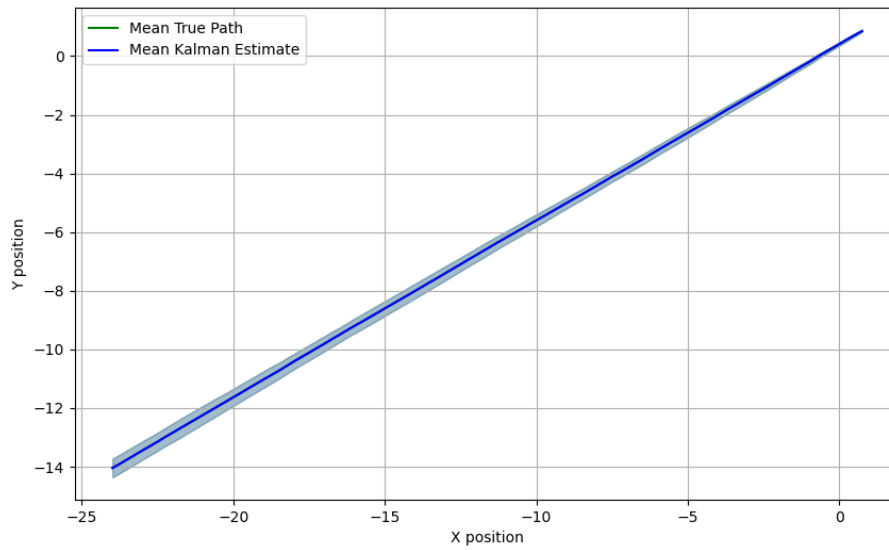


Figure 2: Sample Variance